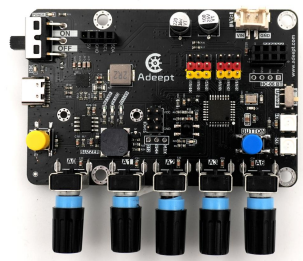


Lesson 3 Control the Servo to Rotate

3.1 Overview

In the field of robot manufacturing, servos are core components. This lesson will help you master the principle of servo control and how to control a servo to rotate to a specified angle.

3.2 Required Components

Components	Quantity	Picture
Adeept Robotic Arm Control Board	1	
Type-C USB Cable	1	
Servo	1	

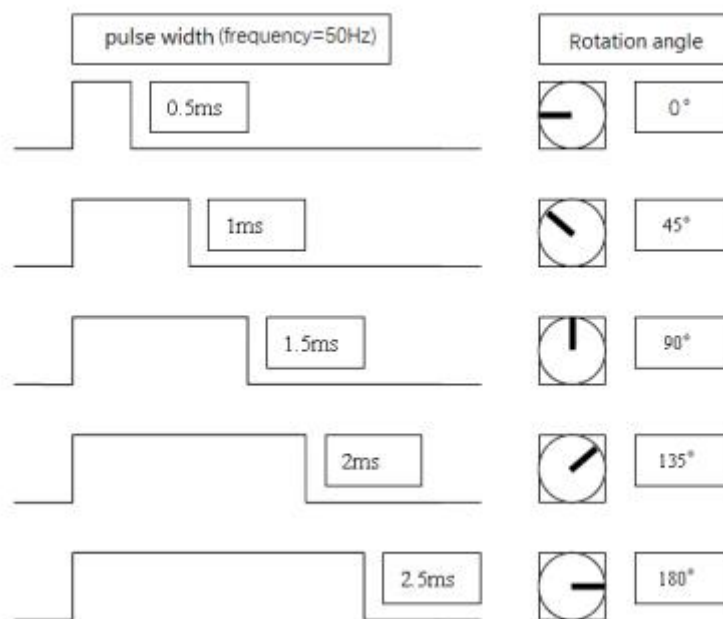
3.3 Principle Introduction

Working principle

The servo operates based on a pulse positioning mechanism. Essentially, when the servo receives a pulse, it rotates by a corresponding angle to achieve displacement. The servo has the ability to send pulses, and for each rotation angle, a specific number of pulses are emitted. This forms a

closed - loop system for pulse reception and transmission, enabling the system to precisely control the motor's rotation and achieve accurate positioning.

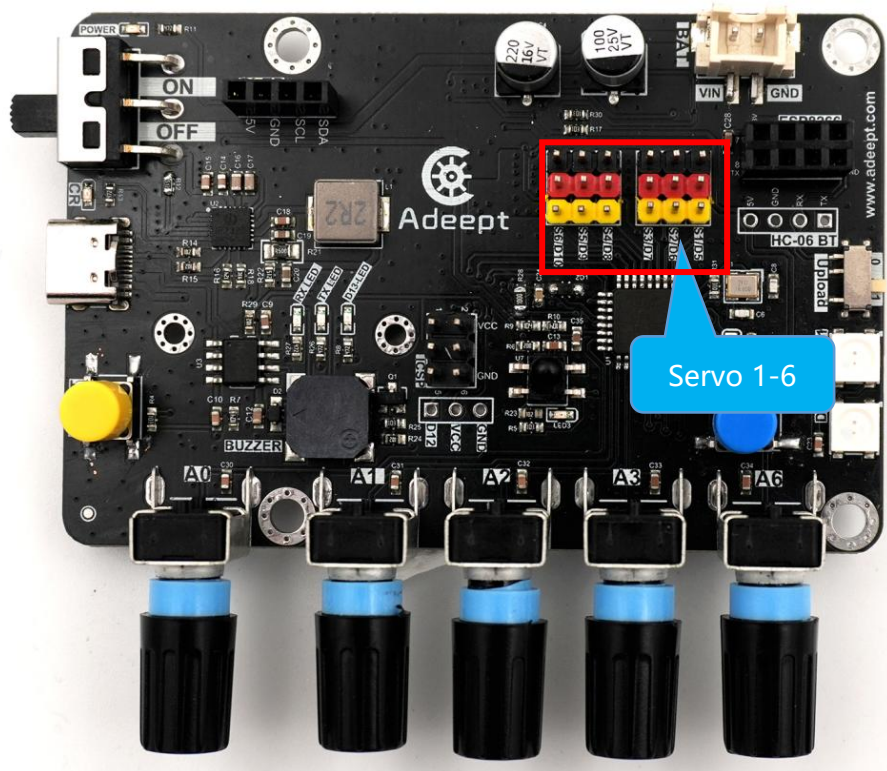
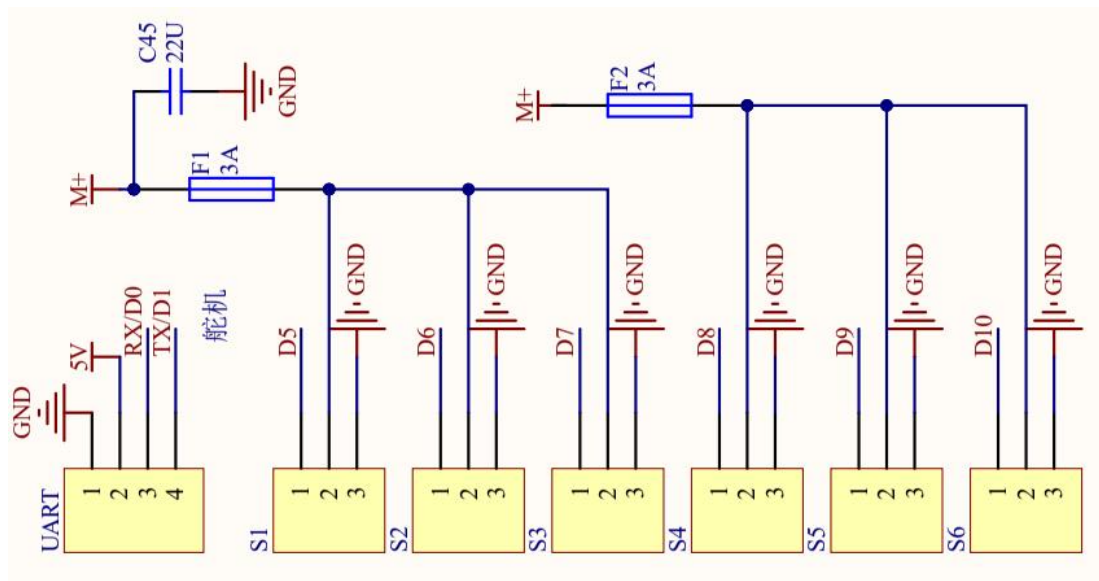
In practical scenarios, the Adeept Robotic Arm Control Board sends PWM (Pulse - Width Modulation) signals to the servo. These signals are processed by the integrated circuit (IC) on the circuit board, which calculates the rotation direction of the driving motor. The rotation is then transmitted to the swing arm through a reduction gear. Simultaneously, the position detector returns a position signal to determine whether the set position has been reached. At a frequency of 50Hz, different pulse widths correspond to different rotation angles: a 0.5ms pulse corresponds to 0°, a 1ms pulse to 45°, a 1.5ms pulse to 90°, a 2ms pulse to 135°, and a 2.5ms pulse to 180°.



Principle of the Write() function

In programming, the write() function is utilized to control the rotation of the servo. For standard servos, this function rotates the servo shaft to the corresponding angular position. For continuous - rotation servos, the write() function can set their rotation speed. A value of 0 represents full - speed rotation in one direction, 180 represents full - speed rotation in the opposite direction, and 90 represents a stop. In this tutorial, we focus on standard servo motors.

3.4 Wiring Diagram



PINS of Adept Robotic Arm Control Board	Servo
S1/D5	Servo1
S2/D6	Servo2

S3/D7	Servo3
S4/D8	Servo4
S5/D9	Servo5
S6/D10	Servo6

3.5 Demonstration

1. Connect your computer and Adeept Robotic Arm Control Board with a USB cable. (Servos require a large operating current, so power the board with batteries and turn on the power switch.)
2. Open "03_Servo" folder in "/Code" , double-click "03_Servo.ino" .

```

03_Servo | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino UNO
03_Servo.ino
1  /*****
2  Website: www.adeept.com
3  E-mail: support@adeept.com
4  *****/
5  #include <Servo.h>
6  #define SERVO_NUM 6
7  Servo servos[SERVO_NUM];
8  uint8_t servoPins[SERVO_NUM] = {5, 6, 7, 8, 9, 10};
9
10 void setup() {
11     for (int i = 0; i < SERVO_NUM; i++) {
12         servos[i].attach(servoPins[i]);
13         servos[i].write(90);
14     }
15     delay(1000);
16 }
17
18 void loop() {
19     servoMoveAll(0);
20     delay(2000);
21     servoMoveAll(180);
22     delay(2000);
23 }

```

3. Select development board and serial port.

Board: **Tools---**>**Board---**>**Arduino AVR Boards---**>**Arduino Uno**

Port: **Tools** ---> **Port**---> **COMx**

Note: The port number will be different in different computers.



4. After opening, click  to upload the code program to the Arduino. If there is no error warning in the console below, it means that the Upload is successful.

```
Output
Sketch uses 2952 bytes (9%) of program storage space. Maximum is 32256 bytes.
Global variables use 82 bytes (4%) of dynamic memory, leaving 1966 bytes for local variables. Maximum is 2048 bytes.
```

5. After the program runs successfully, the onboard servo will rotate in sequence to 0 degrees and 180 degrees, maintaining each angle for 2 second, and so on.

3.6 Code

```
01  /*****
02  Website: www.adeept.com
03  E-mail: support@adeept.com
04  *****/
05  #include <Servo.h>
06  #define SERVO_NUM 6
07  Servo servos[SERVO_NUM];
08  uint8_t servoPins[SERVO_NUM] = {5, 6, 7, 8, 9, 10};
09
10  void setup() {
11      for (int i = 0; i < SERVO_NUM; i++) {
12          servos[i].attach(servoPins[i]);
13          servos[i].write(90);
14      }
15      delay(1000);
16  }
17
18  void loop() {
19      servoMoveAll(0);
20      delay(2000);
21      servoMoveAll(180);
22      delay(2000);
23  }
24
25  void servoMoveAll(int angle) {
26      for (int i = 0; i < SERVO_NUM; i++) {
27          servos[i].write(angle);
28          delay(10);
29      }
30  }
```